PAANIPULSE — Technical Architecture & Cloud Specification

100% Serverless Cloud Infrastructure on AWS • Techfest 2026–27 Campus to Cloud Hackathon

1. Architectural Principles

- **Zero-Idle-Cost Serverless:** Amazon API Gateway, AWS Lambda (Python 3.12), and Amazon DynamoDB.
- **Deterministic Civic Intelligence:** 100-point transparent Community Signal Confidence scoring.
- **Explainable Anomaly Detection:** Diurnal Neighbourhood Water Fingerprint comparison without black-box AI.
- **Privacy-by-Design:** Municipal ward and locality centroids only; zero private home addresses stored.

2. DynamoDB Key Optimization

Partition Key: `localityId` (S) | Sort Key: `timestamp#reportId` (S). Zero table scans; single-digit millisecond query execution.

3. Free Tier Cost Verification

Estimated monthly infrastructure cost: **$0.00 / month** under standard AWS Free Tier.